Inference: Since, the calculated value is much greater than table value i.e., $Z > Z_{\alpha}$, it is highly significant at 5% level of significance. Hence we reject the null hypothesis H_0 and conclude that the advertising campaign was definitely successful in promoting sales.

Example 8.18

The wages of the factory workers are assumed to be normally distributed with mean and variance 25. A random sample of 50 workers gives the total wages equal to ₹ 2,550. Test the hypothesis $\mu = 52$, against the alternative hypothesis $\mu = 49$ at 1% level of significance.

Solution:

Sample size n = 50 workers

Total wages Σx = 2550

Sample mean $\bar{x}$ = $\frac{total\ wages}{n} - \frac{\Sigma x}{n}$

= $\frac{2550}{50} = 51$ units

Population mean μ = 52

Population variance σ^2 = 25

Population SD σ = 5

Under the null hypothesis $H_0 : \mu$ = 52

Against the alternative hypothesis $H_1 : \mu \neq 52$ (Two tail)

Level of significance μ = 0.01

Test statistic $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$

$Z = \frac{51-52}{\frac{5}{\sqrt{50}}} = \frac{-1}{0.7071} = -1.4142$

Since alternative hypothesis is of two tailed test we can take $|Z| = 1.4142$

Critical value at 1% level of significance is $Z_{\alpha/2} = 2.58$

Inference: Since the calculated value is less than table value i.e., $Z < Z_{\alpha/2}$ at 1% level of significance, the null hypothesis H_0 is accepted. Therefore, we conclude that there is no significant difference between the sample mean and population mean μ= 52 and SD σ = 5. Therefore μ = 49 is rejected.

Example 8.19

An ambulance service claims that it takes on the average 8.9 minutes to reach its destination in emergency calls. To check on this claim, the agency which licenses ambulance services has then timed on 50 emergency calls, getting a mean of 9.3 minutes with a standard deviation of 1.6 minutes. What can they conclude at 5% level of significance.

Solution:

Sample size $n = 50$

Sample mean $\bar{x}$ = 9.3 minutes

Sample S.D s = 1.6 minutes

Population mean μ = 8.9 minutes

Null hypothesis $H_0 : \mu = 8.9$

Alternative hypothesis $H_1 : \mu = 8.9$ (Two tail)

Level of significance $\mu = 0.05$

Test statistic

$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$

$Z = \frac{9.3 - 8.9}{\frac{1.6}{\sqrt{50}}} = \frac{0.4}{0.2263} = 1.7676$

Calculated value Z = 1.7676

Critical value at 5% level of significance is $Z_{\alpha/2} = 1.96$

Inference: Since the calculated value is less than table value i.e., $Z < Z_{\alpha/2}$ at 5% level of significance, the null hypothesis is accepted. Therefore we conclude that an ambulance

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